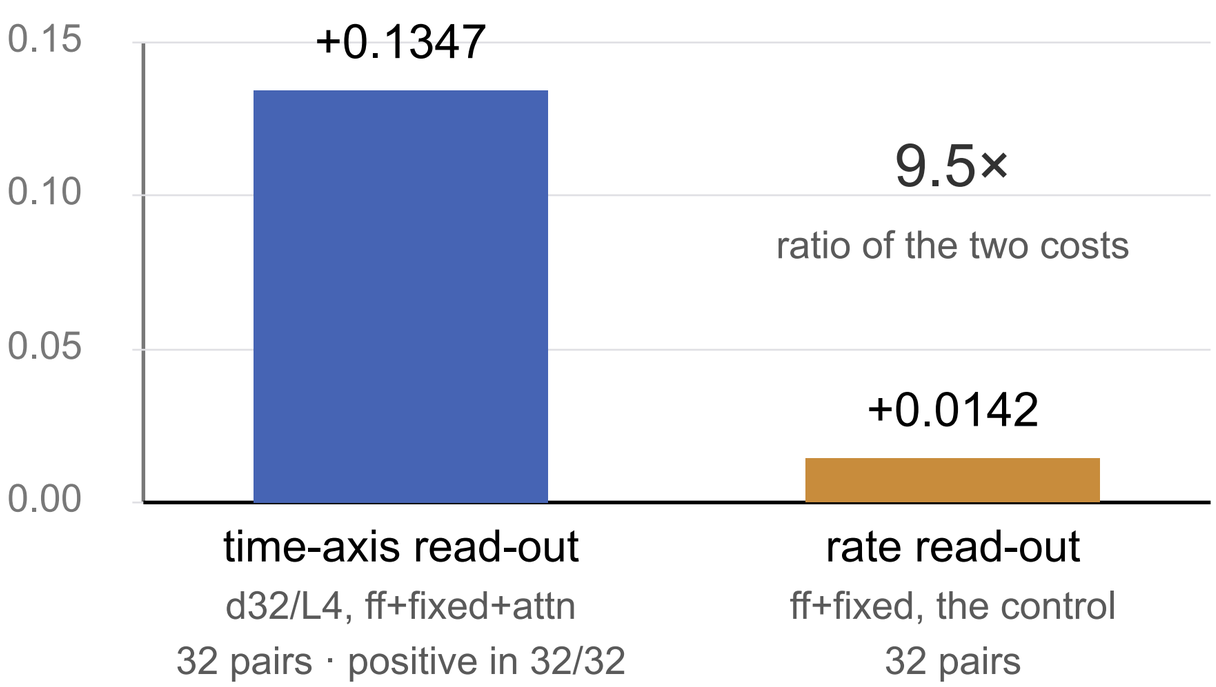
NOT SHOWN HERE, AND NOT CLAIMED: that SHD depends on temporal order. That is established: Cramer et al. 2022 (≤60% on spike-count-only SHD), the Neuromorphic Sequential Arena 2025 (86.48 → 68.51 with temporal processing removed model-side), Yu et al. 2025 (randomised spike times, reversed time).

Panel A: each arm's own shuffle cost. n = 32, seed-paired, h128 / published-2ms / e400.

Bin-shuffling permutes time bins independently per sample in BOTH the training and test splits, so the task itself must be solved without temporal order. Nothing is removed from the model.



Panel B: the same result as the collapse of the advantage over the rate arm.

n = 32 intact +0.1275 → bin-shuffled +0.0070
94.5% of the advantage is contingent on temporal order

n = 12 intact +0.1258 → bin-shuffled +0.0050
96% of the advantage is contingent on temporal order

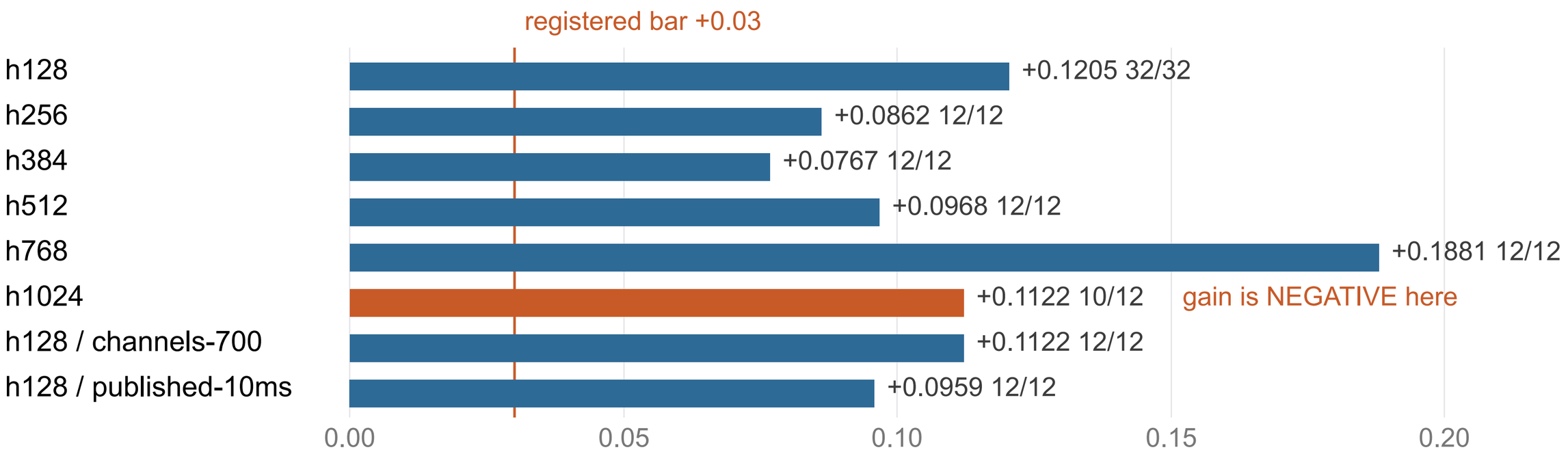
n = 12 is the registration; twenty further seeds move the gain by +0.0017 and the shuffle cost by +0.0010. The larger run confirms the registration; it did not rescue it.

Panel C: the n = 12 measurement, in absolute accuracy. LABELLED n = 12: absolute bin-shuffled means are not published at n = 32.

time-axis read-out	0.8320 → 0.6983	cost +0.1337
rate read-out	0.7062 → 0.6934	cost +0.0128

Attention's cost is positive in 12/12; per seed it falls between +0.0967 and +0.1568, and every w9shf cell passes the temporal audit.

Panel D: the same contrast at eight operating points. n = 12 per point, n = 32 at h128, seed-paired.



The mechanism travels: every point clears the registered bar at BOTH budgets, and coverage is 21 of 21 operating points, NOT every width and NOT every read-out depth: h768/d32l2 is 0.1271 from its d32l4 twin, outside the 0.10 bar.

Its SIZE is not the gain. Spearman rho over the six widths is -0.1430 against a registered bar of +0.829 (NOT MET): h768 carries the smallest positive gain on the ladder (+0.0560) and the largest DiD here (+0.1881).

Points are in ladder order and are deliberately NOT connected and NOT sorted by effect size: either would assert a relationship the rank correlation rejects.